

**Software Design Document**

**AI Career & Skill Guidance System**



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# Table of Content

|                                                  |    |
|--------------------------------------------------|----|
| 1. <b>Summary</b> .....                          | 3  |
| 2. <b>Application Architecture</b> .....         | 4  |
| 2.1 Overall Architecture .....                   | 4  |
| 2.2 Proposed Models .....                        | 5  |
| 2.3 Proposed Views .....                         | 5  |
| 2.4 Controllers and Their Functions .....        | 5  |
| 2.4.1 UserController .....                       | 5  |
| 2.4.2 CareerRecommendationController .....       | 6  |
| 2.4.3 SkillAnalysisController .....              | 6  |
| 2.4.4 ResourceController .....                   | 6  |
| 2.4.5 AutomationController .....                 | 6  |
| 2.5 APIs and External Libraries Used .....       | 7  |
| 2.5.1 AI API (OpenAI API) .....                  | 7  |
| 2.5.2 Integration APIs (Zapier / Make.com) ..... | 7  |
| 2.5.3 Learning Resource APIs .....               | 7  |
| 2.6 External Libraries .....                     | 8  |
| 2.7 UML Diagrams .....                           | 8  |
| 3. <b>Data Design / Data Model Schema</b> .....  | 12 |
| 3.1 Entity Description .....                     | 12 |
| 3.2 Relationships Between Entities .....         | 14 |
| 3.3 Entity Relationship (ER) Diagram .....       | 15 |
| 4. <b>User Interface Design</b> .....            | 17 |
| 4.1 UI Design Philosophy .....                   | 17 |
| • Color Scheme                                   |    |
| • Font Style                                     |    |
| • Design Niche                                   |    |
| • Responsiveness                                 |    |
| 4.2 List of Screens / Pages .....                | 18 |
| 4.2.1 Welcome Screen                             |    |
| 4.2.2 User Input Screen                          |    |
| 4.2.3 Career Recommendation Screen               |    |
| 4.2.4 Career Comparison Screen                   |    |
| 4.2.5 Skill Progress Dashboard                   |    |
| 4.2.6 Learning Resources Screen                  |    |
| 4.3 Input Constraints .....                      | 20 |
| 5. <b>Conclusion</b> .....                       | 21 |

## Summary

The **AI Career & Skill Guidance System** is designed to provide intelligent career guidance to students, beginners, and job seekers who face difficulty in selecting the right career path. Many individuals are confused about which field suits them best and what skills are required to succeed in their chosen career. This system aims to solve that problem by using Artificial Intelligence and automation tools to analyze user information and provide personalized career recommendations.

The system follows a modern design approach that combines AI reasoning, automated workflows, and a simple user interface. Instead of relying on heavy programming, the design focuses on modular components that interact with each other efficiently. The chatbot-based interface allows users to communicate easily with the system, while the AI engine processes user data to generate meaningful insights such as career suggestions, skill gap analysis, and learning roadmaps.

From a design perspective, the system is structured into multiple layers. The presentation layer handles user interaction through chatbot and voice-based communication. The processing layer contains the AI recommendation logic, while the automation layer manages data flow and synchronization. The data layer stores user profiles, skills, and progress records using cloud-based storage. This layered design makes the system easy to understand, scalable, and maintainable.

The design also includes features such as career comparison, skill progress tracking, and visual dashboards. These features help users evaluate different career options and monitor their learning journey over time. Voice interaction is added to improve accessibility and user engagement, making the system more interactive and user-friendly.

Overall, this Software Design Document explains how the **AI Career & Skill Guidance System** is structured internally, how its components communicate with each other, and how the system will be implemented in a practical and efficient manner. The document serves as a blueprint for development and ensures that the system design remains clear, organized, and aligned with project objectives.

## 2. Application Architecture

The **Application Architecture** of the *AI Career & Skill Guidance System* defines the overall structural design of the system and explains how different components interact with each other. The main goal of this architecture is to create a well-organized, scalable, and easy-to-understand system that can deliver accurate career guidance with minimum complexity.

The system is designed using a **layered and modular architecture approach**. Each module has a clearly defined responsibility, which helps in reducing system dependency and improving maintainability. This approach also allows future improvements, such as adding new career fields, learning platforms, or advanced AI features, without affecting the existing system structure.

The architecture focuses on separating user interaction, AI intelligence, automation workflows, and data handling. This separation ensures smooth data flow, efficient processing, and better system performance. The overall design supports both text-based and voice-based interaction, making the system accessible and user-friendly.

The **AI Career & Skill Guidance System** follows a structured, modular, and MVC-inspired architecture. The purpose of this section is to explain the overall design of the system, the architectural pattern being used, the list of Models, Views, and Controllers, the functions handled by each controller, and the APIs or external libraries involved. UML diagrams and coding conventions are also referenced as required by the template.

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### 2.1 Overall Architecture

The system uses a **Layered + MVC-inspired architecture**, which separates the application into three major parts:

- **Model Layer** – Represents system data such as user profiles, skills, careers, and progress.
- **View Layer** – The user interface screens used for interaction.
- **Controller Layer** – Contains business logic, data processing, and communication between the UI and backend logic.

This separation makes the system easy to maintain, easy to modify, and scalable for future improvements. Since the system integrates AI, automation tools, and cloud services, the layered architecture ensures smooth interaction between all components.

## 2.2 Proposed Models

- UserProfile Model
- Career Model
- Skill Model
- LearningResource Model
- ProgressTracking Model

## 2.3 Proposed Views

- User Welcome Screen
  - User Input Screen
  - Career Recommendation Screen
  - Career Comparison Screen
  - Skill Progress Dashboard
  - Learning Resources Screen
- 

## 2.4 Controllers and Their Functions

The various controllers used in the system and the key functions they perform. Controllers act as the intermediary between the user interface and the business logic of the application. Each controller is responsible for handling specific tasks such as managing user data, generating career recommendations, analyzing skills, fetching learning resources, and automating workflows. Understanding these controllers and their functions is crucial for grasping the system's internal operations.

### 2.4.1 UserController

**Purpose:** Handles all user interactions and profile data.

**Main Functions:**

- `collectUserInput()` – Gather user interests, education, skills
- `validateUserData()` – Ensure valid information

- `updateUserProfile()` – Save updated user details

### 2.4.2 CareerRecommendationController

**Purpose:** Generates AI-based career suggestions.

**Main Functions:**

- `analyzeUserProfile()` – Passes user data to AI
- `generateCareerList()` – Retrieves top matching careers
- `compareCareers()` – Compares two careers (growth, skills, learning effort)

### 2.4.3 SkillAnalysisController

**Purpose:** Performs skill gap and progress analysis.

**Main Functions:**

- `identifyRequiredSkills()`
- `calculateSkillGap()`
- `updateProgressStatus()`

### 2.4.4 ResourceController

**Purpose:** Recommends learning courses and tutorials.

**Main Functions:**

- `fetchLearningResources()` – Fetches from YouTube, Coursera, FCC
- `matchResourcesWithSkills()`

### 2.4.5 AutomationController

**Purpose:** Handles automated workflows and data integration.

**Main Functions:**

- `triggerAutomationFlows()`

- `syncUserData()`
  - `syncProgressUpdates()`
- 

## 2.5 APIs and External Libraries Used

Provides an overview of the external APIs and libraries integrated into the system to enhance its capabilities. APIs such as OpenAI's AI API are used to generate intelligent career recommendations, while automation platforms like n8n and Make.com facilitate seamless data synchronization and workflow automation. The system also connects with learning resource APIs to fetch up-to-date educational content. Utilizing these external services allows the system to provide accurate, real-time recommendations and ensures smooth integration with third-party platforms.

### 2.5.1 AI API (OpenAI API)

- Analyzes user input
- Generates personalized career suggestions
- Processes comparison logic

#### Working:

User data → API call → AI-generated recommendation → System displays result

### 2.5.2 Integration APIs (Zapier / Make.com)

- Connects chatbot, sheets, learning platforms
- Automates data flow
- No backend coding needed

#### Working:

Trigger → Automation scenario → Output processing → Return to system

### 2.5.3 Learning Resource APIs

While direct custom APIs are not required, the system fetches resources using:

- YouTube Data API
- Coursera open catalog
- FreeCodeCamp public content

These provide updated learning materials for skills.

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## 2.6 External Libraries

External libraries are used to simplify development and provide advanced features such as voice interaction, data storage, and data visualization. Libraries like Voiceflow are employed for building conversational user interfaces, Google Sheets API for cloud-based data storage, and Looker Studio along with Chart.js for creating interactive dashboards and visual progress reports. Leveraging these libraries reduces development time and enhances user experience.

- **Voiceflow** (UI + chat logic)
- **Google Sheets API** (data storage)
- **Looker Studio** (visual dashboards)
- **Chart.js** (optional progress graphs)

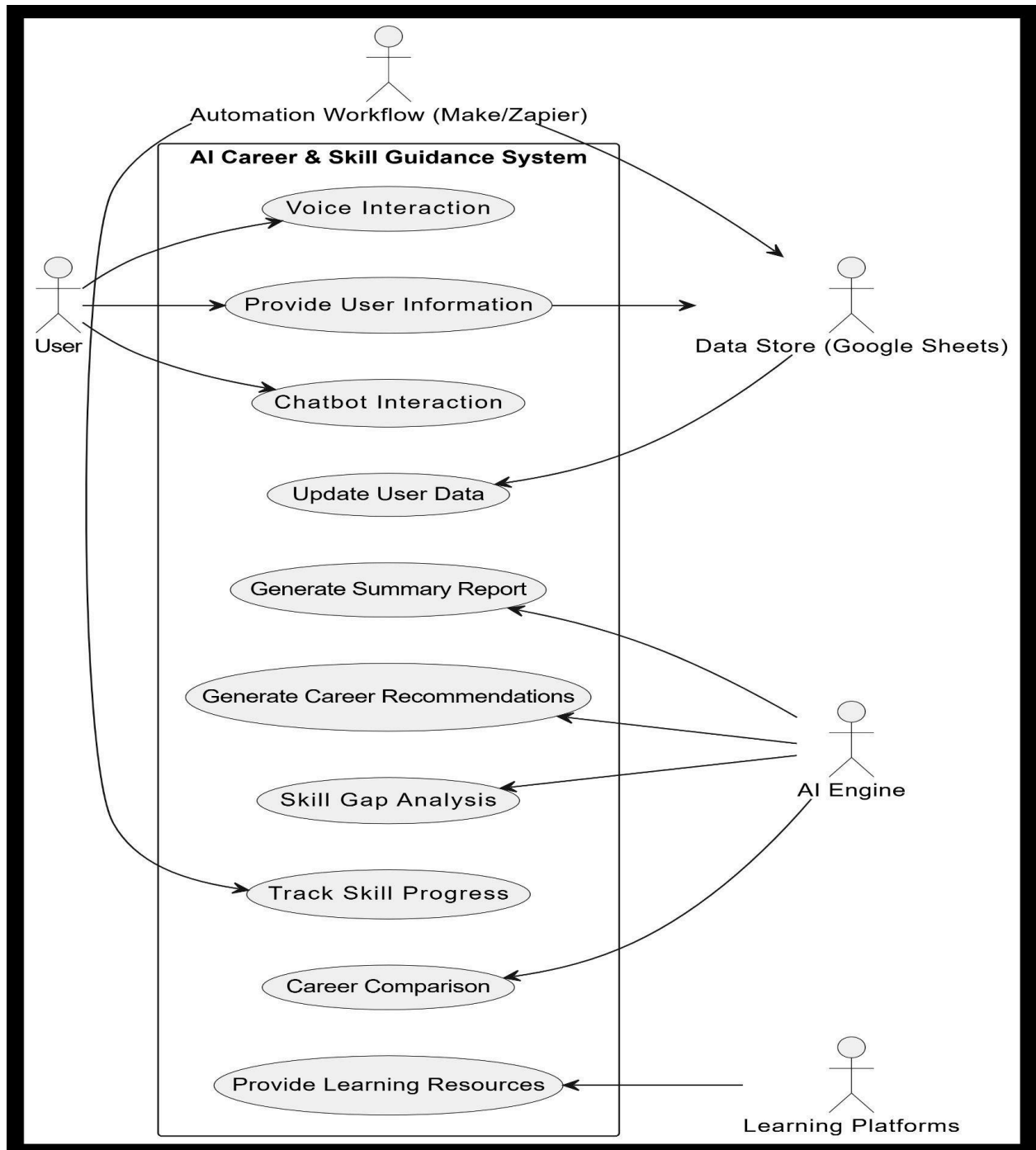
## 2.5. UML Diagrams

Unified Modeling Language (UML) diagrams are visual tools used to represent the structure and behavior of the system. This section includes diagrams such as Use Case Diagrams to depict user interactions, Class Diagrams to show system classes and their relationships, Sequence Diagrams to illustrate the flow of operations, and Entity Relationship Diagrams (ERD) to map the data model. These diagrams help in understanding the system design and facilitate effective communication among developers and stakeholders.

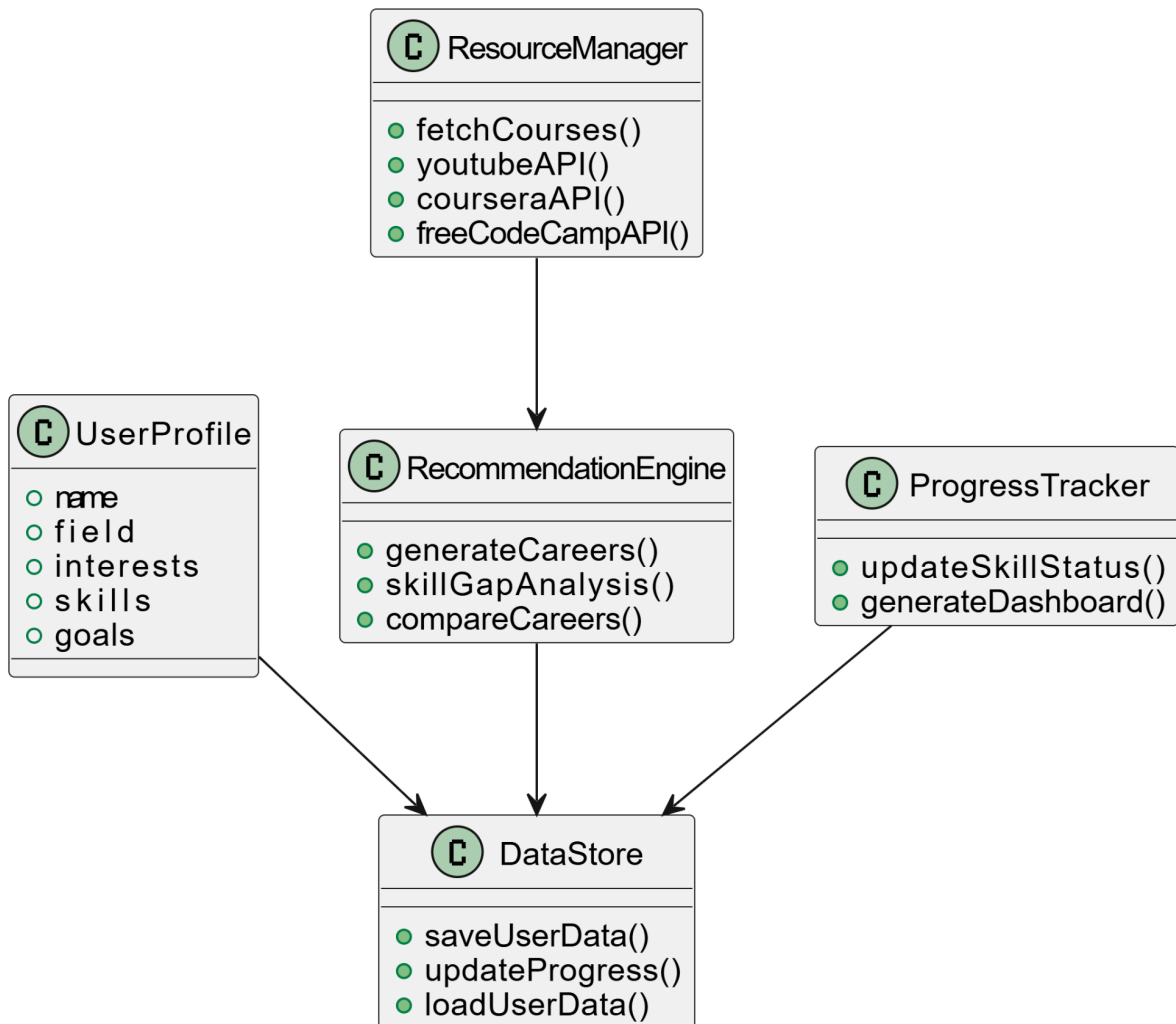
- **Use Case Diagram** – User interactions
- **Class Diagram** – System classes & structure
- **Sequence Diagram** – Flow of recommendation process
- **ER Diagram** – Data model representation



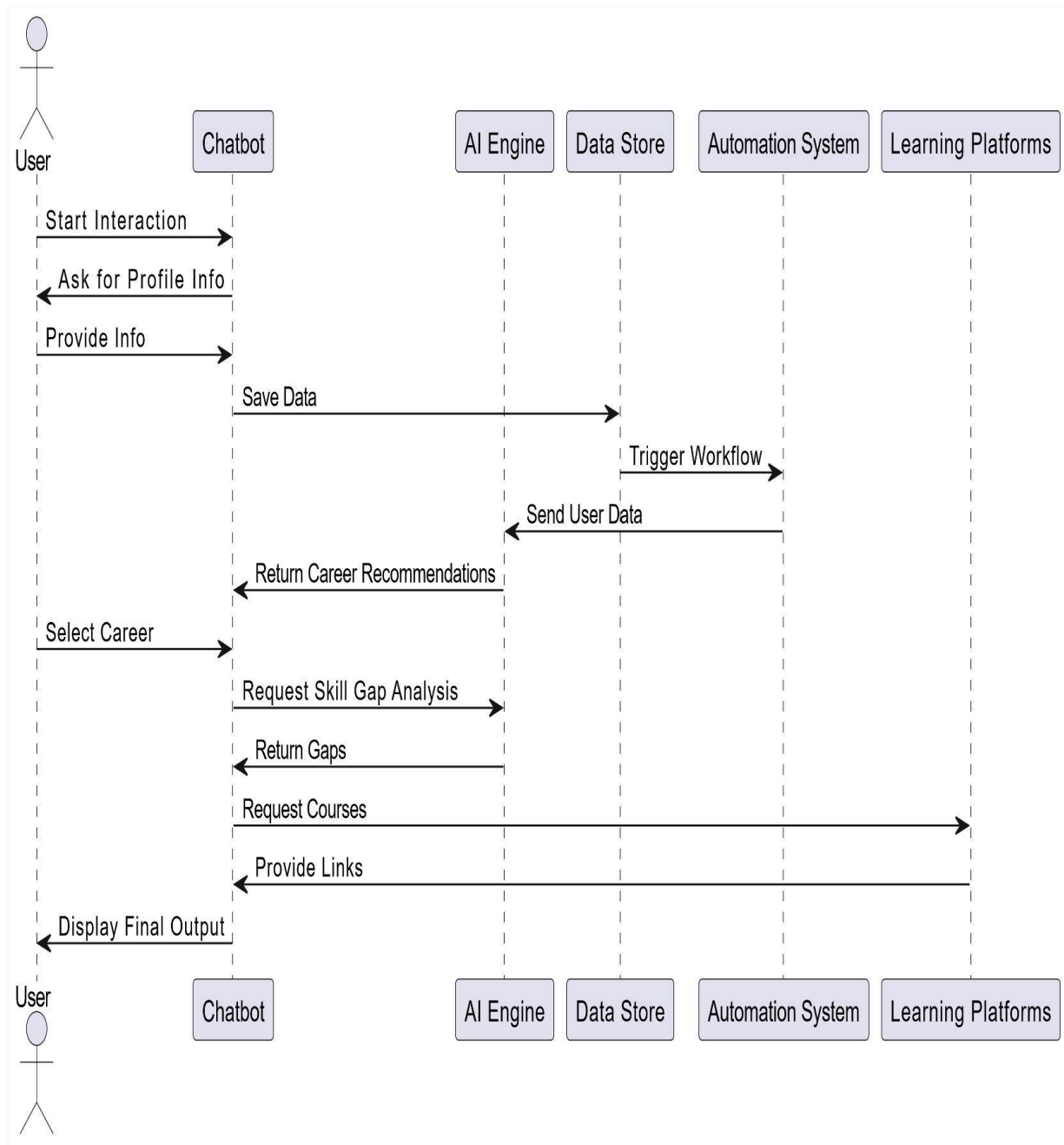
## USE CASE DIAGRAM:



## CLASS DIAGRAM – SYSTEM STRUCTURE:



## SEQUENCE DIAGRAM – USER FLOW :



## 3. Data Design / Data Model Schema

The **Data Design** of the *AI Career & Skill Guidance System* defines how data is structured, stored, and managed within the system. A well-planned data model is essential to ensure that the

system works efficiently, maintains data consistency, and supports future enhancements. This section explains the main data entities, their attributes, and the relationships between them.

The system is designed to handle user-related data, career information, skill records, and progress tracking data. The data model ensures that information flows smoothly between different system components such as the user interface, AI processing layer, and automation layer.

The data design follows a **conceptual and logical modeling approach**, which means the structure is clearly defined even if lightweight or cloud-based storage solutions are used. This approach is suitable for academic projects and modern AI-driven systems.

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### 3.1 Entity Description

An entity represents a real-world object or concept whose data needs to be stored in the system. The following are the main entities used in the AI Career & Skill Guidance System.

#### 3.1.1 User Profile Entity

The **User Profile** entity stores basic information about the user. This entity is essential because all recommendations and analysis are based on user-specific data.

**Attributes include:**

- User ID
- Name
- Educational Background
- Field of Interest
- Existing Skills
- Career Preferences

This entity allows the system to personalize career guidance and track user progress over time.

#### 3.1.2 Skills Entity

The **Skills** entity stores information related to different technical and non-technical skills. These skills are mapped with career paths and learning resources.

**Attributes include:**

- Skill ID
- Skill Name
- Skill Category
- Skill Level (Beginner, Intermediate, Advanced)

This entity helps the AI engine identify skill gaps and suggest relevant skills for career growth.

### **3.1.3 Career Entity**

The **Career** entity represents different career paths available in the system. Each career is associated with a set of required skills and learning recommendations.

**Attributes include:**

- Career ID
- Career Title
- Career Description
- Required Skills
- Growth Potential

This entity supports career comparison and recommendation features.

### **3.1.4 Learning Resources Entity**

The **Learning Resources** entity stores information about educational materials such as courses, tutorials, and videos.

**Attributes include:**

- Resource ID
- Resource Title
- Platform Name
- Resource Type
- Access Link

This entity allows the system to suggest learning paths and guide users toward skill development.

### 3.1.5 Progress Tracking Entity

The **Progress Tracking** entity records user learning activities and progress status. This entity plays a key role in monitoring skill development.

**Attributes include:**

- Progress ID
- User ID
- Skill ID
- Completion Status
- Last Updated

Using this entity, the system can show progress reports and adjust recommendations accordingly.

---

## 3.2 Relationships Between Entities

Relationships define how different entities are connected to each other. In this system, relationships are designed to support personalization and intelligent analysis.

- A **User Profile** can have multiple **Skills**
- A **Career** is associated with multiple **Skills**
- A **User Profile** can explore multiple **Careers**
- **Learning Resources** are linked to both **Skills** and **Careers**
- **Progress Tracking** connects **User Profiles** with **Skills**

These relationships allow the system to analyze user data effectively and generate meaningful career guidance.

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### 3.3 Entity Relationship (ER) Diagram

The **Entity Relationship (ER) Diagram** visually represents the data model of the AI Career & Skill Guidance System. It shows entities, their attributes, and the relationships between them.

The ER diagram helps in:

- Understanding the overall data structure
- Identifying data dependencies
- Supporting future database implementation

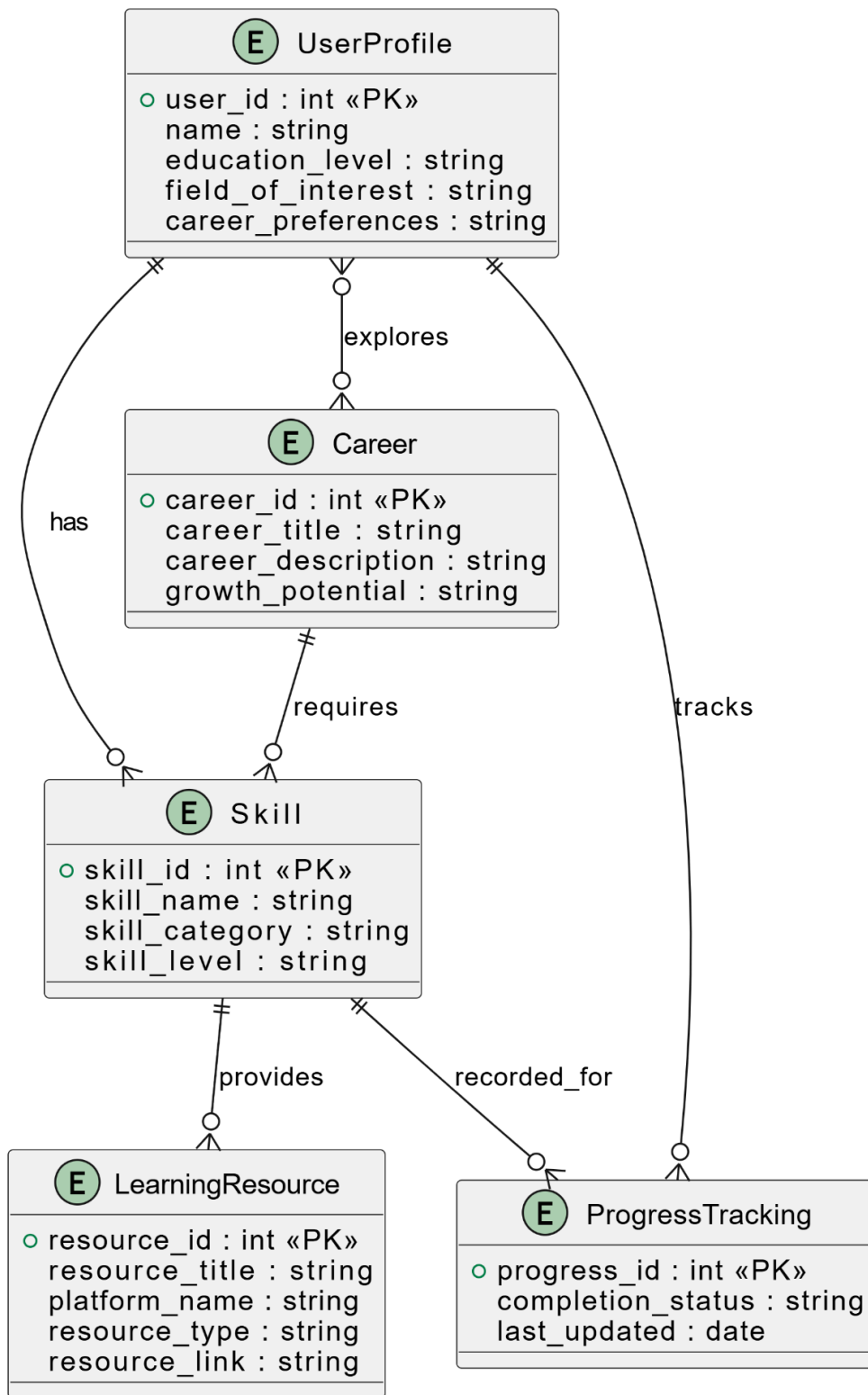
The diagram includes entities such as User Profile, Skills, Career, Learning Resources, and Progress Tracking, along with their relationships.

The Entity Relationship (ER) Diagram represents the data structure of the AI Career & Skill Guidance System. It shows key entities including UserProfile, Skill, Career, LearningResource, and ProgressTracking along with their relationships. The diagram supports personalized career recommendations, career comparison, and skill progress tracking by clearly defining how user data, skills, and learning resources are connected. This structured data model ensures efficient data management and supports future system scalability.

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**Entity Relationship (ER) Diagram:**

## ER Diagram - AI Career & Skill Guidance System





## 4. User Interface Design

The User Interface (UI) of the **AI Career & Skill Guidance System** is designed to be simple, modern, and user-friendly. The main goal of the UI is to ensure that users can easily enter their information, view personalized recommendations, and navigate between screens without confusion. The overall design emphasizes clarity, accessibility, and visual balance.

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### 4.1 UI Design Philosophy

#### Color Scheme

The system follows a clean and soft color palette to maintain readability and professionalism:

- **Primary Colors:** Blue and Teal (for trust, technology, and learning)
- **Secondary Colors:** Light Gray and White (for minimal and clean backgrounds)
- **Highlight Color:** Green (for progress indicators and positive actions)

#### Font Style

- **Primary Font:** Poppins / Roboto
- **Reason:** Modern, readable, and suitable for digital dashboards.
- **Font Weight Usage:**
  - Headings → Semi-bold
  - Subheadings → Medium
  - Description/Text → Regular

#### Design Niche

The design follows:

- **Minimalist UI approach**
- **Dashboard-style layout**
- **Card-based components** for clarity

- **Soft rounded corners** to give a modern look
- **Consistent spacing and alignment**

## **Responsiveness**

The interface is designed to work on:

- Desktop / Laptop (Primary target)
- Tablet (Medium screen compatible)
- Mobile (Responsive layout with stacked cards)

The system scales content automatically using flex/grid layout.

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## **4.2 List of Screens / Pages**

### **4.2.1. Welcome Screen**

**Purpose:** Introduces the system and navigates user to start.

**Controls:**

- “Get Started” Button
- Data Shown:** System introduction text.
- 

### **4.2.2. User Input Screen**

**Purpose:** Collects user data for AI analysis.

**Controls:**

- Textbox (Name, Education, Interests)
- Dropdown (Skill Level, Career Interest)
- Multi-select options (Existing skills)
- “Submit Details” Button

**Input Constraints:**

- Text fields accept only alphabetical or properly formatted input
  - Dropdown selections are mandatory
  - Minimum one skill or interest required
- 

**4.2.3. Career Recommendation Screen**

**Purpose:** Shows AI-generated career matches.

**Data Displayed:**

- List of top recommended careers
- Matching percentage
- Required skills
- Career growth rating

**Controls:**

- “View Details” Button
  - “Compare Careers” Button
- 

**4.2.4. Career Comparison Screen**

**Purpose:** Compares two selected careers side by side.

**Data Displayed:**

- Skill requirements
- Salary range
- Growth rate
- Learning difficulty
- Job roles

**Controls:**

- Dropdown to select first career
- Dropdown to select second career
- “Compare Now” Button

#### **4.2.5. Skill Progress Dashboard**

**Purpose:** Shows user's learning progress.

**Data Displayed:**

- Skill improvement chart
- Learning hours completed
- Recommended next steps

**Controls:**

- Progress Chart (Chart.js visualization)
- “Update Progress” Button

#### **4.2.6. Learning Resources Screen**

**Purpose:** Displays courses fetched from YouTube, Coursera, FreeCodeCamp.

**Data Displayed:**

- Course title
- Duration
- Difficulty level
- Direct link

**Controls:**

- Search Bar
- Filters (Beginner / Intermediate / Advanced)
- “Start Learning” Button

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## 4.3 Input Constraints

To ensure data reliability and AI accuracy:

- **Name field:** Letters only
- **Education field:** Must be descriptive (min 3 characters)
- **Skills field:** At least one skill required
- **Career selection:** Mandatory in comparison screen
- **Progress update:** Numeric values only
- **Search bar:** No special characters except letters/numbers

## 5. Conclusion

The AI Career & Skill Guidance System is designed to provide intelligent, personalized, and accessible career guidance for students and job seekers. The system integrates modern AI technologies, automation tools, and cloud-based data storage to deliver accurate career recommendations, skill gap analysis, and learning resources.

The software design focuses on modularity, scalability, and ease of use, ensuring that the system can be enhanced in the future without significant restructuring. By combining a user-friendly interface with powerful backend AI processing and automation workflows, the system aims to help users make informed career decisions and effectively track their skill development.

This Software Design Document provides a clear blueprint for implementation, supporting both academic evaluation and practical application. It ensures that all components are well-defined, interoperable, and aligned with project objectives.

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